

Inclusive AI Fashion Stylist with Measurement and Color-Blind Safety Analysis

Abstract—Digital clothing selection can be challenging because users need to identify garment colors, assess outfit compatibility, determine suitable sizes, and choose clothing that matches their preferences. These challenges can be particularly difficult for individuals with color vision deficiency, who may struggle to distinguish visually similar colors and patterns. This study develops an integrated artificial intelligence-based fashion assistance framework that supports users with and without color vision deficiency. The framework combines clothing color analysis, color vision deficiency-aware perception, outfit compatibility assessment, virtual body measurement, clothing size recommendation, and personalized fashion guidance. It identifies dominant and multiple garment colors, analyzes their distribution, and generates perception-aware representations under different color vision conditions. Outfit compatibility is assessed using perceptual color relationships and machine learning classification. Body landmarks extracted from full-body images are used to estimate body measurements and recommend appropriate clothing sizes using a user-provided height reference. The framework also incorporates catalogue-constrained fashion recommendation to match user requirements with available clothing attributes and provide personalized styling suggestions. Experimental evaluation demonstrates reliable performance across color analysis, compatibility assessment, body landmark detection, and size recommendation under controlled image-capture conditions. Overall, the findings demonstrate the feasibility of integrating accessibility, measurement, sizing, and personalized recommendation into a unified platform for inclusive clothing selection.

Keywords—*Color Vision Deficiency, Clothing Color Analysis, Outfit Compatibility, Clothing Size Recommendation*

I. INTRODUCTION

Garments are chosen by using digital media which can be used to obtain information about the looks of the garment, its size, and its features; however, digital media does not offer an equally convenient and personalized experience for everyone. People who have a color vision deficiency (CVD) face problems in recognizing the colors of the garment, in differentiating between two visually similar colors, and in determining whether a set of garments would match considering their own unique perception of color. The problem is specifically associated with multicolored and

patterned garments. In case of normal consumers, there are standard size charts available which may not fully reflect individual physical features. Existing research has primarily addressed these challenges as separate problems. Color recognition and CVD simulation have largely focused on general image analysis rather than fashion-specific applications, while clothing-matching approaches have commonly emphasized color similarity without providing an interpretable assessment of overall outfit compatibility. Image-based body measurement methods may require specialized equipment or be limited by population characteristics, and existing fashion recommendation approaches may not adequately account for the constraints of real-world product catalogues. Consequently, an integrated approach capable of addressing accessibility, sizing, compatibility, and personalized fashion selection remains insufficiently explored.

The developed framework integrates four complementary functions within a unified fashion assistance platform: CVD-aware clothing color analysis for identifying dominant and multiple garment colors and representing their appearance under different CVD conditions; outfit compatibility assessment for evaluating the relationship between garment colors and identifying potential CVD-related color distinguishability issues; virtual body measurement and clothing-size recommendation for estimating body dimensions from full-body images and determining suitable size categories; and adaptive fashion recommendation and styling for interpreting user requirements and retrieving catalogue-constrained clothing options with personalized styling guidance. By integrating these functions through a common image-processing and catalogue-based architecture, the framework provides a comprehensive approach to accessible and personalized clothing selection for users with diverse visual, physical, and preference-related requirements.

II. RELATED WORK

A. AI-Based CVD-Aware Clothing Colour Analysis

Color detection in clothing, CVD simulation, and clothing pattern improvement are important research topics in accessible fashion image analysis. Previous works have used machine learning techniques and pixel-wise approaches for

clothing color detection and extraction [1] [2]. Wong et al. have designed a shirt-color recognition system for people with color vision deficiencies, which has been limited mostly to predefined colors [1]. In contrast, traditional color extraction methods focus on dominant or representative colors [2].

To address the shortcomings mentioned above, the system has utilized MobileNetV2 for the primary clothing color detection based on RGB- and Lab-based analysis and KMeans clustering to detect more than one color along with their percentage of distribution. Color vision deficiencies have been extensively investigated to make images more accessible for people with color blindness. Brettel et al. have designed a color transformation system for the simulation of dichromatic vision, whereas Machado et al. have proposed an LMS model for the simulation of protanopia, deuteranopia, and tritanopia [3] [4]. Other studies have explored color enhancement and daltonization [5], while deep-learning-based CVD compensation has also been investigated [6]. However, these methods mainly target general images rather than fashion-specific color perception. Similarly, CNN-based approaches have demonstrated effectiveness in textile texture and pattern classification and color analysis [7], but they do not specifically enhance pattern visibility for CVD users. Therefore, the proposed system integrates clothing color recognition, CVD simulation, and pattern enhancement to support clearer garment perception and more informed, accessible clothing selection.

B. AI-Based Virtual Size Recommendation

Body landmark detection and anthropometric measurement are critical aspects in the context of the image-based body measurement and fashion analysis. Some existing works examined computer vision methods for the identification of body landmarks and the estimation of anthropometric measurements in the context of clothes [8], [9]. However, some of the examined approaches are limited to specific population types, likewise young Asian women [9], while a Kinect-based anthropometric system was validated only on male subjects and was not tested on women [10]. This may affect the applicability of these systems in terms of different genders and age categories. Other works have used machine learning techniques to predict the body measurements for clothing-size prediction purposes, but most of them used 3D body scanning systems that require special equipment [11], [12]. Also, some commercial AI body analysis products tried to generate style, workout, or health-related suggestions based on the provided image [13]. In contrast, the proposed system estimates 17 key points of full body posture for both males and females regardless of the targeted age category, uses the height reference supplied by the user instead of depth sensors for estimating body measurements, recommends clothing sizes from XS to 4XL, and includes an AI-based Body Appearance & Wellness Guide based on gender, height, and body proportions.

C. Multimodal and Context-Aware Fashion Recommendation

Visual ranking methods show that product appearance is valuable side information for recommendation, motivating the use of catalog attributes as a grounding signal rather than relying solely on interaction history [14]. Context-aware recommendation research demonstrates that situational variables, such as an occasion or event, can meaningfully alter what is relevant to a user, supporting the design choice to

weight occasion only when it is explicitly supplied rather than inferring it. [15] Large multimodal language models such as Qwen3-VL provide the language and multimodal reasoning capabilities used to convert free-form text into structured, catalog-constrained fields in this system [16]. Learned ranking objectives such as Bayesian personalized ranking (BPR) offer an alternative to the weighted, rule-based scoring used here, and are identified as a direction for future comparison [17].

D. Clothing Colour Matching and Compatibility Assessment

Beyond color recognition and CVD simulation, outfit color compatibility assessment has mainly relied on perceptual color-difference metrics. CIEDE2000 is widely adopted because it closely agrees with human perceptual judgements of color difference [18]. Separately, clothes-matching systems for blind and color-blind users have used camera-based color comparison to determine whether two garments' colors are alike, handling both plain and multi-colored items [19]. However, these approaches either report a single color distance value without translating it into an interpretable compatibility category, or emphasize structural and textural matching rather than combining multi-color perceptual distance with a trained classification model. To address this gap, the proposed system extracts a weighted multi-color palette for each garment, computes CIEDE2000 distance across every color pair, and combines the resulting distance statistics with an ensemble of a Random Forest and a Neural Network to classify the pairing as Good, Moderate, or Poor. In parallel, each garment's dominant color is checked independently against known CVD color-confusion pairs, so users receive both a general match-quality score and a CVD-specific readability check for the top and bottom garments separately.

III. METHODOLOGY

A. AI-Based CVD-Aware Clothing Colour Analysis

Clothing images were subjected to a preprocessing stage involving RGB conversion, rescaling, background removal, and CLAHE enhancement. Dominant color categorization was performed using an ImageNet pre-trained EfficientNetB2 model incorporating a color attention network mechanism, trained using categorical cross-entropy loss with the Adam optimizer within a two-step transfer learning approach. For multi-colored garments, the K-Means clustering technique was applied to RGB pixels extracted from the clothing region. Cluster centroids were classified based on the RGB color dataset, and color proportions were computed from their distribution. YCrCb-based skin-pixel exclusion was further applied to remove likely skin regions from the garment mask, thereby reducing interference from the face and hands during color analysis.

A conditional U-Net employing MobileNetV2 as the encoder was developed to simulate three common types of color vision deficiency (CVD): protanopia, deuteranopia, and tritanopia. The conditional CVD image constitutes the model output, generated from an RGB clothing image and a categorical CVD type as input. Visual features are extracted via the MobileNetV2 encoder, while the U-Net decoder generates the CVD image through skip connections to preserve garment borders and colors. For vivid and multicolored garments, the CVD transformation is applied while preserving the spatial color information of individual garment regions, thereby

enabling distinct colors and patterns within the same garment to be represented under the selected CVD condition.

Garment backgrounds were first removed, and the remaining pixels were used to generate pseudo-ground-truth masks representing the principal color regions. A U-Net model with a MobileNetV2 encoder was subsequently employed for pixel-level color segmentation. The encoder extracts visual features, while the decoder reconstructs spatial color regions using skip connections that preserve garment boundaries and fine details. Background pixels were excluded from loss and evaluation calculations. The model was evaluated using masked pixel accuracy, IoU, precision, recall, and F1-score. CLAHE and edge-based enhancement were further applied to improve the visibility of garment patterns and fine details, particularly in regions where color differences may be difficult for CVD users to distinguish.

B. Clothing Color Matching and Compatibility Assessment

Following background removal, each garment's foreground pixels are clustered using K-Means into up to three dominant colors, each associated with a coverage percentage. The CIEDE2000 color-difference metric is computed between every color pair across the top and bottom garments' palettes, weighted by each color's coverage, yielding a weighted average, minimum, maximum, and range of color distance. These statistics, together with the dominant RGB values and color counts of each garment, constitute a twelve-dimensional feature vector. A Random Forest classifier and a feed-forward Neural Network are trained on this feature vector to predict a Good, Moderate, or Poor match category, with training labels auto-generated from a rule requiring both a high average color distance and a low distance range for a "Good" label. During inference, the probabilities produced by both models are averaged to obtain the final match category and a confidence score. Independently of the matching score, each garment's dominant color is mapped to a named color category by hue, saturation, and brightness, and checked against a set of color pairs known to be difficult to distinguish under protanopia, deuteranopia, and tritanopia. This check is performed separately for the top and bottom garments, allowing the system to flag which specific item, if any, may be difficult for the user to distinguish under their color vision deficiency type.

C. AI-Based Virtual Size Recommendation

A custom dataset of full-body images was manually compiled for this study and annotated in CVAT using 17 human body key points [14]. The annotated dataset was used to train a custom YOLO26l pose estimation model [15] using Python in Google Collab. Anaconda was also used as implemented in Python via Google Colab. Anaconda was used as a local Python distribution and environment-management platform to organise project dependencies and install compatible packages. OpenCV was employed for image processing, and NumPy for numerical calculations. Detected key points were used to calculate shoulder, hip, and body-height measurements based on pixel distances. Ethical clearance was obtained prior to the validation phase for the collection of participant data.

The detected pixel-based body measurements and pose-derived features are extracted and provided as inputs to the trained machine-learning model to predict the appropriate clothing size. Euclidean distance was used to calculate

distances between relevant detected key points. The resulting shoulder and hip measurements, together with body height, were compared with a universal size chart containing standardized measurement ranges from XS to 4XL to determine the recommended size. Before the validation part applied for the Ethical Clearance and then each participants were informed and within their perdition full body images and then their actual shoulder, hip, and height measurements were manually recorded.

The estimated body measurements and predicted clothing size were provided to an external API as inputs for generating personalized Body Appearance & Wellness Guidance. The API was integrated as a separate component after the body measurement and clothing-size prediction stages. Based on the predicted size category, from XS to 4XL, the system generates suitable general appearance and wellness suggestions. These suggestions are intended to provide supportive guidance according to each size category and do not constitute clothing recommendations or medical advice. This component extends the system beyond size prediction while keeping the measurement and clothing-size prediction processes independent from the API.

D. System Architecture and Recommendation Methodology

The component is organised as three services operating behind a FastAPI interface. First, a Qwen3-VL:8B model served through Ollama receives the customer text and a constrained prompt containing valid catalogue categories, returning JSON fields for occasion, preferred colours, article types, styles, materials, patterns, sleeve types, neck types, other requirements, and optional price bounds. Extracted values are validated against the product vocabulary; an absent occasion remains null. This structured interface leverages the model's language and multimodal reasoning capabilities without permitting free-form generations to bypass catalogue constraints

Second, the recommendation engine filters the 15-attribute product table by gender and available size, which constitute mandatory constraints. Each remaining product receives a field score based on exact (1.0), substring (0.7), word-level (0.4), or article-type synonym (0.6–0.9) agreement. When an occasion is specified, weights are assigned as follows: article type 0.20, occasion 0.30, colour 0.20, style 0.15, pattern 0.05, material 0.05, sleeve 0.025, and neckline 0.025. In the absence of an occasion, the redistributed weights become 0.30, 0.25, 0.20, 0.10, 0.05, 0.05, and 0.05 for article type, colour, style, pattern, material, sleeve, and neckline, respectively. Description relevance is also considered prior to ranking. Formally, $S(p,q)$ denotes the weighted sum of field matches for product p and query q . The engine returns the top five candidates with $S \geq 0.35$, accompanied by a natural language explanation of the matched attributes.

Third, following product selection, the stylist component combines the chosen item's type, colour, occasion, style, material, pattern, and price with the original request. Qwen3VL produces a structured summary covering accessories, footwear, complementary colours, layering, and related items. Advice is occasion-specific only when an occasion was supplied; otherwise, it remains versatile. The service avoids body-related judgements and defaults to baseline attribute ranking if model inference is unavailable. The complete application employs a Next.js and React interface, MongoDB for user-level data storage, and a CSV-based product catalogue

IV. RESULT

A. AI-Based CVD-Aware Clothing Color Analysis

The EfficientNetB2 model achieved 83% accuracy and a balanced F1-score of 0.83 across all classes. While visually distinct colors like white (0.97) and yellow (0.94) scored near-perfectly, performance dropped for similar colors like brown (0.75) and purple (0.75). Confusion matrix analysis indicates errors primarily occur among perceptually close pairs (e.g., brown-orange, purple-blue). Future work will target these ambiguities via a color-difference loss function and lighting-augmented training.

	precision	recall	f1-score	predicted support
black	0.80	0.75	0.78	60
blue	0.84	0.87	0.85	60
brown	0.76	0.73	0.75	60
green	0.94	0.98	0.85	60
orange	0.79	0.92	0.85	75
pink	0.78	0.83	0.81	60
purple	0.73	0.77	0.75	60
red	0.81	0.78	0.80	60
white	0.97	0.97	0.97	60
yellow	0.97	0.92	0.94	61
accuracy			0.83	616
macro avg	0.84	0.83	0.83	616
weighted avg	0.84	0.83	0.83	616

Fig.IV.1. Per-class classification performance for plain colors

The model's training and validation accuracy curves exhibited stable convergence with no significant overfitting, achieving a final validation accuracy of 83.13%. This confirms the model's generalization capability on unseen data. For multi-color garment analysis, K-Means clustering of RGB pixels mapped to a color dataset enabled quantitative pixel-wise percentage estimation of each garment's color composition.

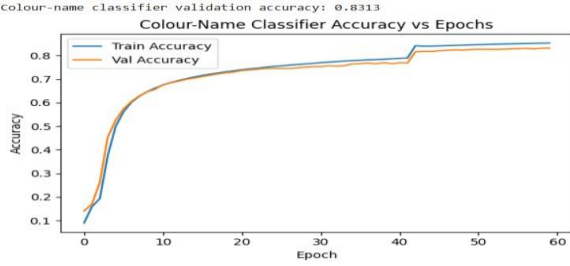


Fig.IV.2. Model accuracy convergence across training epochs multi colors.

The CVD simulation model was trained using a two-phase schedule to ensure stable convergence. The model was trained across two phases (Table II), achieving stable convergence with validation accuracy plateauing at 0.54. Evaluation for protanopia, deuteranopia, and tritanopia using accuracy, confirmed reliable color transformation with preserved garment structure.

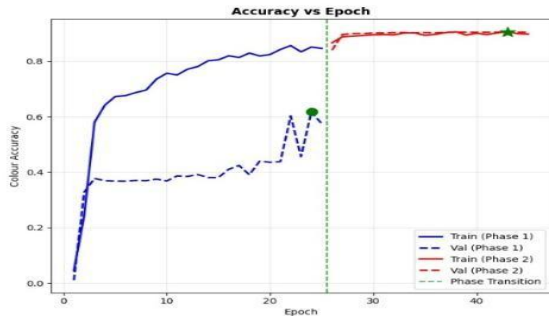


Fig.3. Two-phase training and validation accuracy progression of the CVD simulation model across epochs

B. Clothing Colour Matching and CVD Readability Evaluation

The color-matching pipeline was evaluated on a dataset of 3,714 candidate top-bottom garment pairs, of which 3,708 pairs (99.8%) were processed successfully; Weighted average CIEDE2000 color distance and its associated range were used to auto-generate match-quality labels, yielding 2,876 Moderate pairs (77.6%), 648 Poor pairs (17.5%), and 184 Good pairs (5.0%).

On a held-out test set of 742 pairs (80/20 stratified split), the Random Forest classifier achieved 99.9% accuracy, with precision, recall, and F1-score of 1.00 for both the Moderate and Poor categories and 1.00, 0.97, and 0.99, respectively for the good category (n = 37). Good pair predicted as Moderate, as shown in the confusion matrix (rows = actual, columns = predicted): [36, 1, 0], [0, 575, 0], [0, 0, 130]] for the Good, Moderate, and Poor classes.

Feature importance analysis of the trained Random Forest, shown in Figure —, indicates that the weighted-average color distance (weighted_avg_de) accounted for approximately 65% of total importance, followed by the minimum, maximum, and range of color distance (min_de, max_de, derange), which together accounted for nearly all of the remaining importance. The dominant-color RGB channels and color-count features for both garments contributed negligibly to the classification decision, indicating that the model relies almost entirely on the aggregate color-distance statistics rather than the specific hues involved.

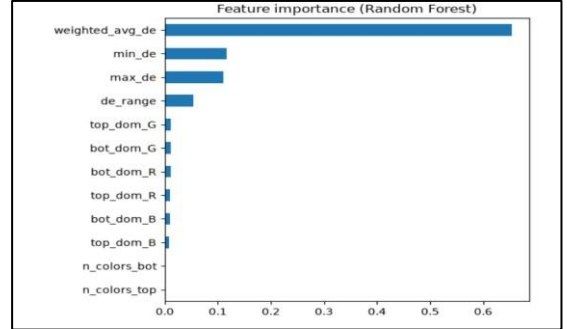


Fig.4. Random Forest Feature Importance for Color-Matching Classification

The feed-forward Neural Network was trained for up to 60 epochs and reached a peak validation accuracy of 94.9% at epoch 50, with validation accuracy fluctuating between approximately 87% and 95% over the later epochs. This slower and less stable convergence relative to the Random Forest is consistent with a tree-based model more readily representing the threshold-based structure of the autogenerated labels. Sample color-pair extractions, together with their assigned match category and weighted average ΔE score, are illustrated in Figure —.



Fig.5. Sample Multi-Color Top/Bottom Pairs with Weighted Average ΔE and Match Category

Because the Good/Moderate/Poor labels are generated directly from the same color-distance statistics used as model features, the reported accuracies primarily validate that both classifiers reliably reproduce the labelling rule; correspondence between this rule and human perceptual judgements of outfit matching, together with validation of the per-garment CVD confusion-pair flags against known colorconfusion tables for protanopia, deuteranopia, and tritanopia, remains part of ongoing evaluation.

C. AI-Based Virtual Size Recommendation

The key point detection results showed that pose model correctly identified the main body landmarks required for measurement extraction. It detected important points such as the shoulders, elbows, wrists, hips, knees, and ankles, enabling the system to estimate body dimensions from input images. The detected landmarks formed a clear and recognizable human-body structure. Overall, the results indicate that the model provided sufficiently accurate key point locations to support body-measurement extraction and clothing-size prediction. Training curves illustrating precision, recall, mAP@0.5, and mAP@0.5:0.95 for bounding box (B) and pose (P) predictions over 100 epochs, demonstrating the convergence and training stability of the proposed pose estimation model.

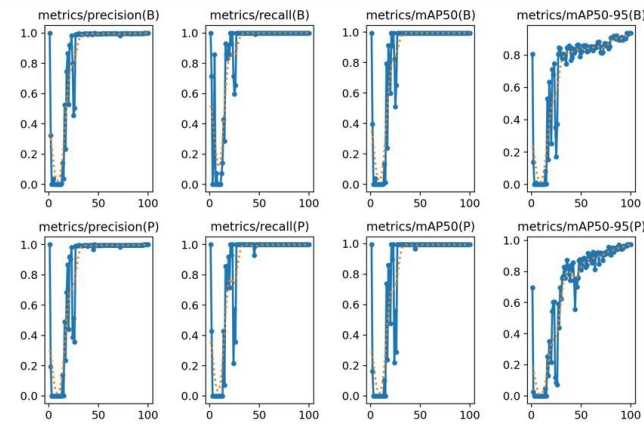


Fig.6. Training results of the proposed pose estimation model obtained after model training in the Google Colab

The proposed solution achieved 80% accuracy in clothing size prediction during the validation part. In most tested cases, the system generated the correct clothing-size category when the input image was clear and captured under suitable conditions. However, there are some incorrect predictions because of the poor lightning and sometimes the distance between the camera and the participant changed. These findings indicate that the approach can provide reliable size recommendations in controlled image-capture conditions, while improvements in image normalization and measurement estimation could further improve prediction consistency. The trained pose estimation model successfully detects the human body and identifies the annotated keypoints across various test images.

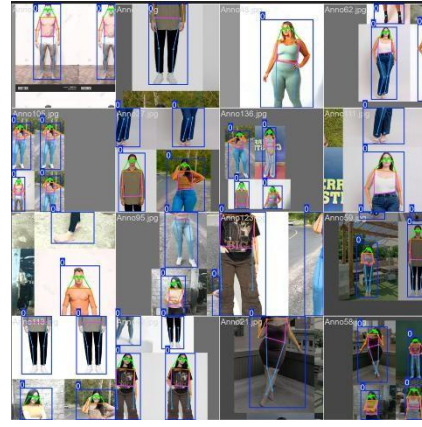


Fig.7. Test results showing detected body keypoints

The Body Appearance & Wellness Guide successfully generated personalized guidance from the estimated body measurements and proportions. The output provided general body appearance and wellness information rather than clothing suggestions. This additional component demonstrates that the system can provide value beyond clothing-size prediction by presenting measurement-based guidance in a user-friendly form. The guidance was treated as supportive information and was not used as a medical assessment or diagnosis.

D. Recommendation Pipeline Validation and Performance Evaluation

The prototype evaluation reported in the source implementation yielded approximately 90% accuracy for explicit requirement extraction, 95.2% for occasion extraction, and 98% correct null detection when no occasion was mentioned. Field-level extraction was strongest for color (94.1%) and article type (92.3%), while material extraction was lower (85.4%). The reported occasion-invention rate was 2.1%, directly measuring the principal safety objective of preserving unspecified context.

For retrieval, reported overall Precision@5, Recall@5, and NDCG@5 were 0.72, 0.68, and 0.75. Precision@5 was 0.78 for requests with an occasion and 0.65 without one; the occasion-aware configuration was reported as an 18.5% improvement over the baseline. Requirement analysis required roughly 5-10 s, recommendation ranking 1-2 s, and stylist generation 8-15 s. These values suggest that deterministic filtering and scoring are fast, whereas model calls dominate latency. They should nevertheless be interpreted as preliminary implementation results because the source does not specify an independently reproducible test split, annotation protocol, confidence intervals, or statistical significance. Larger and more diverse catalogs, controlled ablations of weight redistribution, and user studies are required before generalizing the gains. Future work should also compare learned ranking objectives such as Bayesian personalized ranking [4], add image-text embeddings, and calibrate explanations against observed user choices. The proposed requirement extraction component achieved strong performance, with the highest accuracy for occasion (95.2%), followed by color (94.1%) and article type (92.3%). Overall, the results demonstrate that Qwen3-VL:8B effectively

extracts structured fashion attributes from natural-language requirements for recommendation

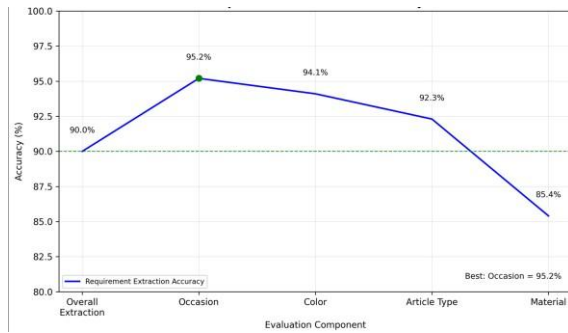


Fig.8. Requirement Extraction Accuracy Across Fashion Attributes

CONCLUSION

This study developed an AI-assisted fashion assistance system integrating four modules to support accessible and personalized clothing selection for both CVD and non-CVD users. The framework provides clothing color analysis, outfit compatibility assessment, virtual size recommendation, and adaptive fashion recommendation with catalogue-based styling guidance. The results demonstrate the potential of integrating these functions within a unified fashion assistance platform. Future work will focus on perceptual validation of CVD-related color predictions, improving the robustness of size estimation, and further evaluating the quality and relevance of generated recommendations.

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